

1 SRT Divider

1.1 Synthesis Reports

1.1.1 Timing Summary

Setup Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 1	∞	54	55	80	rst	Qu[7]	28.960	8.702	20.258	∞	input port clock			0.000
 Path 2	∞	52	53	80	rst	Qu[6]	28.065	8.492	19.573	∞	input port clock			0.000
 Path 3	∞	49	50	80	rst	Qu[5]	26.719	8.177	18.542	∞	input port clock			0.000
 Path 4	∞	46	47	80	rst	Qu[4]	25.340	7.862	17.478	∞	input port clock			0.000
 Path 5	∞	42	43	80	rst	R[7]	24.457	7.462	16.995	∞	input port clock			0.000
 Path 6	∞	42	43	80	rst	R[5]	24.286	7.462	16.824	∞	input port clock			0.000
 Path 7	∞	42	43	80	rst	R[6]	24.286	7.462	16.824	∞	input port clock			0.000
 Path 8	∞	42	43	80	rst	R[4]	24.275	7.462	16.813	∞	input port clock			0.000
 Path 9	∞	43	44	80	rst	Qu[3]	23.994	7.547	16.447	∞	input port clock			0.000
 Path 10	∞	41	42	80	rst	R[3]	23.823	7.357	16.466	∞	input port clock			0.000

Hold Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 11	∞	3	4	5	divisor[7]	e	1.992	1.352	0.640	-∞	input port clock			0.000
 Path 12	∞	3	4	16	divisor[0]	R[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 13	∞	3	4	16	divisor[1]	R[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 14	∞	3	4	14	divisor[2]	R[2]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 15	∞	3	4	16	divisor[4]	R[4]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 16	∞	3	4	15	divisor[5]	R[5]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 17	∞	3	4	15	divisor[5]	R[6]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 18	∞	3	4	5	divisor[7]	R[7]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 19	∞	4	5	15	divisor[5]	Qu[0]	2.207	1.399	0.808	-∞	input port clock			0.000
 Path 20	∞	4	5	18	divisor[3]	R[3]	2.207	1.399	0.808	-∞	input port clock			0.000

1.1.2 Utilization Report

Name 	Slice LUTs (134600)	Bonded IOB (400)
 srt	204	36

1.1.3 Power Report

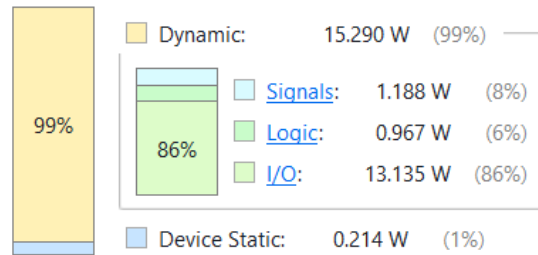
Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: **15.504 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **54.0°C**
Thermal Margin 31.0°C (16.3 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



1.2 Implementation Reports

1.2.1 Timing Summary

Setup Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
Path 1	∞	51	50	80	rst	Qu[7]	36.240	8.623	27.617	∞	input port clock			0.000
Path 2	∞	50	49	80	rst	Qu[6]	35.992	8.877	27.115	∞	input port clock			0.000
Path 3	∞	47	46	80	rst	Qu[5]	33.960	8.208	25.752	∞	input port clock			0.000
Path 4	∞	44	43	80	rst	Qu[4]	32.079	7.877	24.202	∞	input port clock			0.000
Path 5	∞	41	40	80	rst	R[7]	31.851	7.925	23.926	∞	input port clock			0.000
Path 6	∞	41	40	80	rst	R[5]	31.687	7.904	23.783	∞	input port clock			0.000
Path 7	∞	41	40	80	rst	R[6]	31.581	7.900	23.681	∞	input port clock			0.000
Path 8	∞	41	40	80	rst	R[4]	31.460	7.937	23.524	∞	input port clock			0.000
Path 9	∞	40	39	80	rst	R[3]	31.195	7.825	23.370	∞	input port clock			0.000
Path 10	∞	42	41	80	rst	Qu[3]	30.952	8.016	22.936	∞	input port clock			0.000

Hold Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
Path 11	∞	3	2	13	divisor[6]	R[7]	2.453	1.401	1.052	-∞	input port clock			0.000
Path 12	∞	3	2	16	divisor[1]	e	2.459	1.447	1.012	-∞	input port clock			0.000
Path 13	∞	3	2	16	divisor[0]	R[0]	2.530	1.389	1.142	-∞	input port clock			0.000
Path 14	∞	3	2	16	divisor[1]	R[1]	2.537	1.408	1.129	-∞	input port clock			0.000
Path 15	∞	4	3	16	divisor[1]	R[2]	2.595	1.447	1.148	-∞	input port clock			0.000
Path 16	∞	3	2	15	divisor[5]	R[6]	2.606	1.372	1.234	-∞	input port clock			0.000
Path 17	∞	3	2	16	divisor[4]	R[4]	2.625	1.418	1.206	-∞	input port clock			0.000
Path 18	∞	4	3	18	divisor[3]	R[5]	2.672	1.407	1.265	-∞	input port clock			0.000
Path 19	∞	4	3	16	divisor[0]	Qu[2]	2.734	1.436	1.298	-∞	input port clock			0.000
Path 20	∞	4	3	16	divisor[0]	Qu[1]	2.765	1.424	1.341	-∞	input port clock			0.000

1.2.2 Utilization Report

Name  1	Slice LUTs (134600)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (400)
N srt	204	57	204	36

1.2.3 Power Report

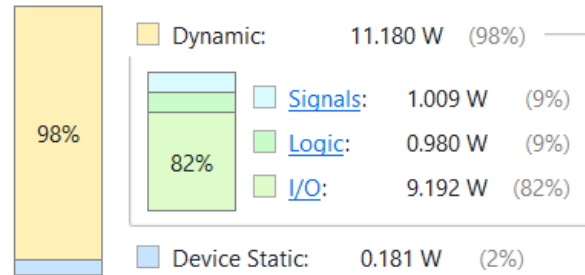
Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **11.361 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **46.2°C**
Thermal Margin 38.8°C (20.4 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



2 Non-Restoring Divider

2.1 Synthesis Reports

2.1.1 Timing Summary

Setup Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 1	∞	72	73	123	clk	Qu[4]	42.378	10.672	31.706	∞	input port clock			0.000
 Path 2	∞	71	72	123	clk	Qu[3]	41.934	10.567	31.367	∞	input port clock			0.000
 Path 3	∞	68	69	123	clk	R[6]	40.621	10.272	30.349	∞	input port clock			0.000
 Path 4	∞	68	69	123	clk	R[7]	40.621	10.272	30.349	∞	input port clock			0.000
 Path 5	∞	67	68	123	clk	R[5]	40.158	10.167	29.991	∞	input port clock			0.000
 Path 6	∞	66	67	123	clk	R[4]	39.695	10.062	29.633	∞	input port clock			0.000
 Path 7	∞	65	66	123	clk	R[3]	39.232	9.957	29.275	∞	input port clock			0.000
 Path 8	∞	65	66	123	clk	Qu[2]	38.919	9.937	28.982	∞	input port clock			0.000
 Path 9	∞	64	65	123	clk	R[2]	38.769	9.852	28.917	∞	input port clock			0.000
 Path 10	∞	64	65	123	clk	Qu[1]	38.475	9.832	28.643	∞	input port clock			0.000

Hold Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 11	∞	3	4	123	clk	Qu[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 12	∞	3	4	60	rst	Qu[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 13	∞	3	4	123	clk	Qu[2]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 14	∞	3	4	60	rst	Qu[3]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 15	∞	3	4	123	clk	Qu[4]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 16	∞	3	4	123	clk	Qu[5]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 17	∞	3	4	60	rst	Qu[6]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 18	∞	3	4	8	divisor[0]	R[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 19	∞	3	4	7	divisor[1]	R[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 20	∞	3	4	9	divisor[2]	R[2]	1.994	1.354	0.640	-∞	input port clock			0.000

2.1.2 Utilization Report

Name 	Slice LUTs (134600)	Bonded IOB (400)
 NON	142	37

2.1.3 Power Report

Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power:

9.598 W

Design Power Budget:

Not Specified

Process:

typical

Power Budget Margin:

N/A

Junction Temperature:

42.9°C

Thermal Margin

42.1°C (22.2 W)

Ambient Temperature

25.0 °C

Effective θ JA

1.9°C/W

Power supplied to off-chip devices

0 W

Confidence level

Low

Launch Power Constraint Advisor to find and fix invalid switching activity

On-Chip Power

98%

Dynamic: 9.429 W (98%)

85%

Signals: 0.738 W (8%)

Logic: 0.667 W (7%)

I/O: 8.024 W (85%)

Device Static: 0.169 W (2%)

2.2 Implementation Reports

2.2.1 Timing Summary

Setup Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
Path 1	∞	70	69	123	clk	Qu[4]	53.979	12.215	41.765	∞	input port clock			0.000
Path 2	∞	69	68	123	clk	Qu[3]	53.186	12.110	41.076	∞	input port clock			0.000
Path 3	∞	64	63	123	clk	Qu[2]	49.739	11.595	38.144	∞	input port clock			0.000
Path 4	∞	67	66	123	clk	R[7]	49.486	12.276	37.210	∞	input port clock			0.000
Path 5	∞	67	66	123	clk	R[6]	49.334	12.251	37.083	∞	input port clock			0.000
Path 6	∞	66	65	123	clk	R[5]	48.999	12.150	36.849	∞	input port clock			0.000
Path 7	∞	63	62	123	clk	Qu[1]	48.785	11.478	37.307	∞	input port clock			0.000
Path 8	∞	65	64	123	clk	R[4]	48.549	11.895	36.654	∞	input port clock			0.000
Path 9	∞	64	63	123	clk	R[3]	47.892	11.600	36.292	∞	input port clock			0.000
Path 10	∞	63	62	123	clk	R[2]	47.292	11.678	35.614	∞	input port clock			0.000

Hold Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
Path 11	∞	3	2	7	divisor[1]	R[1]	2.347	1.413	0.934	-∞	input port clock			0.000
Path 12	∞	3	2	7	divisor[4]	R[4]	2.443	1.408	1.035	-∞	input port clock			0.000
Path 13	∞	3	2	123	clk	Qu[6]	2.511	1.407	1.104	-∞	input port clock			0.000
Path 14	∞	3	2	10	divisor[7]	e	2.516	1.379	1.138	-∞	input port clock			0.000
Path 15	∞	3	2	8	divisor[3]	R[3]	2.530	1.412	1.118	-∞	input port clock			0.000
Path 16	∞	4	3	7	divisor[4]	R[5]	2.616	1.486	1.131	-∞	input port clock			0.000
Path 17	∞	3	2	6	divisor[7]	R[7]	2.621	1.384	1.237	-∞	input port clock			0.000
Path 18	∞	3	2	6	divisor[6]	R[6]	2.626	1.373	1.253	-∞	input port clock			0.000
Path 19	∞	3	2	9	divisor[2]	R[2]	2.650	1.471	1.179	-∞	input port clock			0.000
Path 20	∞	3	2	8	divisor[0]	R[0]	2.660	1.388	1.272	-∞	input port clock			0.000

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2.2.2 Utilization Report

Name  1	Slice LUTs (134600)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (400)
N NON	142	40	142	37

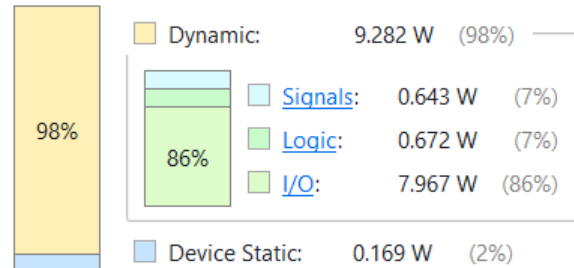
2.2.3 Power Report

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **9.451 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **42.7°C**
Thermal Margin 42.3°C (22.3 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



3 Restoring Divider

3.1 Synthesis Reports

3.1.1 Timing Summary

Setup Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 1	∞	59	60	90	rst	Qu[1]	35.775	9.267	26.508	∞	input port clock			0.000
 Path 2	∞	58	59	90	rst	Qu[0]	35.323	9.162	26.161	∞	input port clock			0.000
 Path 3	∞	55	56	90	rst	r[6]	33.757	8.847	24.910	∞	input port clock			0.000
 Path 4	∞	53	54	90	rst	r[7]	32.843	8.637	24.206	∞	input port clock			0.000
 Path 5	∞	44	45	90	rst	r[2]	27.635	7.692	19.943	∞	input port clock			0.000
 Path 6	∞	42	43	90	rst	r[0]	26.175	7.482	18.693	∞	input port clock			0.000
 Path 7	∞	34	35	90	rst	r[5]	21.654	6.622	15.032	∞	input port clock			0.000
 Path 8	∞	32	33	90	rst	r[4]	20.527	6.412	14.115	∞	input port clock			0.000
 Path 9	∞	29	30	90	rst	r[3]	19.115	6.097	13.018	∞	input port clock			0.000
 Path 10	∞	19	20	90	rst	valid	13.600	5.027	8.573	∞	input port clock			0.000

Hold Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 11	∞	3	4	90	rst	Qu[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 12	∞	3	4	90	rst	Qu[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 13	∞	3	4	90	rst	Qu[2]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 14	∞	3	4	116	clk	Qu[3]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 15	∞	3	4	90	rst	Qu[4]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 16	∞	3	4	90	rst	Qu[5]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 17	∞	3	4	90	rst	Qu[6]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 18	∞	3	4	90	rst	Qu[7]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 19	∞	3	4	90	rst	r[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 20	∞	3	4	90	rst	r[1]	1.994	1.354	0.640	-∞	input port clock			0.000

3.1.2 Utilization Report

Name 	Slice LUTs (134600)	Bonded IOB (400)
 divider	139	35

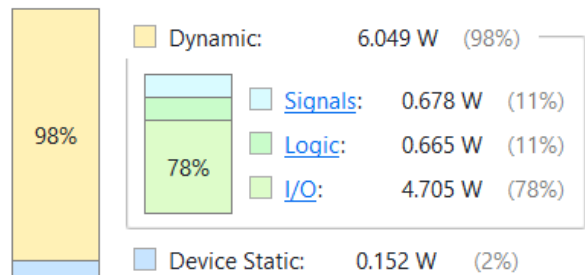
3.1.3 Power Report

Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: **6.201 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **36.6°C**
Thermal Margin 48.4°C (25.6 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



3.2 Implementation Reports

3.2.1 Timing Summary

Setup Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 1	∞	59	58	116	clk	Qu[1]	45.544	10.504	35.039	∞	input port clock			0.000
↳ Path 2	∞	58	57	116	clk	Qu[0]	44.829	10.365	34.464	∞	input port clock			0.000
↳ Path 3	∞	55	54	116	clk	r[6]	43.853	10.092	33.761	∞	input port clock			0.000
↳ Path 4	∞	53	52	116	clk	r[7]	43.253	10.053	33.201	∞	input port clock			0.000
↳ Path 5	∞	44	43	116	clk	r[2]	36.526	8.744	27.782	∞	input port clock			0.000
↳ Path 6	∞	42	41	116	clk	r[0]	34.722	8.351	26.371	∞	input port clock			0.000
↳ Path 7	∞	34	33	116	clk	r[5]	29.623	7.356	22.267	∞	input port clock			0.000
↳ Path 8	∞	32	31	116	clk	r[4]	27.976	6.959	21.017	∞	input port clock			0.000
↳ Path 9	∞	29	28	116	clk	r[3]	25.999	6.654	19.345	∞	input port clock			0.000
↳ Path 10	∞	18	17	116	clk	r[1]	18.376	5.304	13.072	∞	input port clock			0.000

Hold Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 11	∞	3	2	90	rst	Qu[0]	2.072	1.319	0.754	-∞	input port clock			0.000
↳ Path 12	∞	3	2	90	rst	Qu[2]	2.109	1.352	0.757	-∞	input port clock			0.000
↳ Path 13	∞	3	2	90	rst	Qu[1]	2.113	1.353	0.760	-∞	input port clock			0.000
↳ Path 14	∞	3	2	116	clk	Qu[4]	2.300	1.360	0.940	-∞	input port clock			0.000
↳ Path 15	∞	3	2	90	rst	Qu[5]	2.369	1.337	1.032	-∞	input port clock			0.000
↳ Path 16	∞	3	2	90	rst	Qu[6]	2.369	1.336	1.033	-∞	input port clock			0.000
↳ Path 17	∞	3	2	116	clk	Qu[7]	2.415	1.383	1.031	-∞	input port clock			0.000
↳ Path 18	∞	4	3	5	D[7]	r[7]	2.575	1.506	1.070	-∞	input port clock			0.000
↳ Path 19	∞	3	2	116	clk	r[4]	2.576	1.396	1.180	-∞	input port clock			0.000
↳ Path 20	∞	4	3	90	rst	Qu[3]	2.599	1.440	1.159	-∞	input port clock			0.000

3.2.2 Utilization Report

Name 	Slice LUTs (134600)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (400)
N divider	138	40	138	35

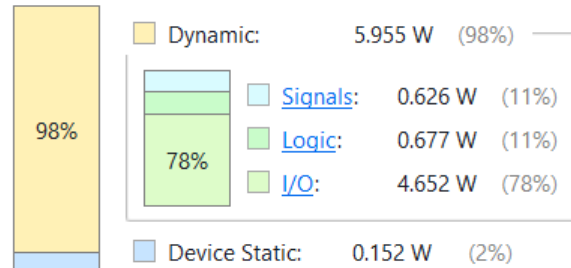
3.2.3 Power Report

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **6.106 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **36.4°C**
Thermal Margin 48.6°C (25.7 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



4 Newton-Raphson Divider

4.1 Synthesis Reports

4.1.1 Timing Summary

Setup Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 1	∞	104	105	31	divisor[1]	quotient[1]	66.307	14.072	52.235	∞	input port clock			0.000
 Path 2	∞	104	105	31	divisor[1]	quotient[3]	66.307	14.072	52.235	∞	input port clock			0.000
 Path 3	∞	104	105	31	divisor[1]	quotient[5]	66.307	14.072	52.235	∞	input port clock			0.000
 Path 4	∞	104	105	31	divisor[1]	quotient[0]	66.066	14.072	51.994	∞	input port clock			0.000
 Path 5	∞	104	105	31	divisor[1]	quotient[2]	66.066	14.072	51.994	∞	input port clock			0.000
 Path 6	∞	104	105	31	divisor[1]	quotient[4]	66.066	14.072	51.994	∞	input port clock			0.000
 Path 7	∞	104	105	31	divisor[1]	quotient[6]	66.066	14.072	51.994	∞	input port clock			0.000
 Path 8	∞	104	105	31	divisor[1]	quotient[7]	66.062	14.092	51.970	∞	input port clock			0.000

Hold Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 9	∞	3	4	38	divisor[7]	quotient[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 10	∞	3	4	38	divisor[7]	quotient[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 11	∞	3	4	38	divisor[7]	quotient[2]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 12	∞	3	4	38	divisor[7]	quotient[4]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 13	∞	3	4	38	divisor[7]	quotient[5]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 14	∞	3	4	15	divisor[3]	quotient[6]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 15	∞	4	5	38	divisor[7]	quotient[3]	2.211	1.399	0.812	-∞	input port clock			0.000
 Path 16	∞	4	5	38	divisor[7]	quotient[7]	2.332	1.398	0.934	-∞	input port clock			0.000

4.1.2 Utilization Report

Name 	Slice LUTs (134600)	Bonded IOB (400)
 divider	291	27

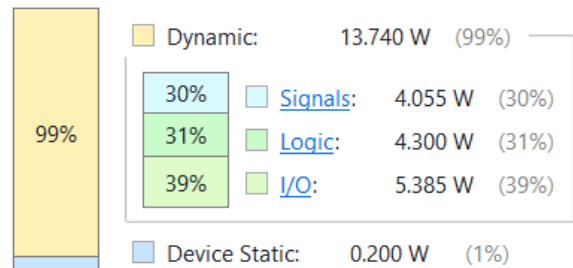
4.1.3 Power Report

Summary

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: **13.941 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **51.1°C**
Thermal Margin 33.9°C (17.9 W)
Ambient Temperature 25.0 °C
Effective θ_{JA} 1.9°C/W
Power supplied to off-chip devices 0 W
Confidence level [Low](#)
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



4.2 Implementation Reports

4.2.1 Timing Summary

Setup Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 1	∞	104	103	31	divisor[2]	quotient[3]	84.233	15.734	68.499	∞	input port clock			0.000
↳ Path 2	∞	104	103	31	divisor[2]	quotient[6]	84.206	15.758	68.449	∞	input port clock			0.000
↳ Path 3	∞	104	103	31	divisor[2]	quotient[7]	84.203	15.757	68.446	∞	input port clock			0.000
↳ Path 4	∞	104	103	31	divisor[2]	quotient[2]	84.136	15.739	68.397	∞	input port clock			0.000
↳ Path 5	∞	104	103	31	divisor[2]	quotient[0]	83.963	15.764	68.199	∞	input port clock			0.000
↳ Path 6	∞	104	103	31	divisor[2]	quotient[4]	83.951	15.747	68.204	∞	input port clock			0.000
↳ Path 7	∞	104	103	31	divisor[2]	quotient[1]	83.889	15.762	68.127	∞	input port clock			0.000
↳ Path 8	∞	104	103	31	divisor[2]	quotient[5]	83.764	15.747	68.017	∞	input port clock			0.000

Hold Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 9	∞	4	3	38	divisor[7]	quotient[3]	2.388	1.405	0.983	-∞	input port clock			0.000
↳ Path 10	∞	3	2	38	divisor[7]	quotient[0]	2.451	1.389	1.061	-∞	input port clock			0.000
↳ Path 11	∞	3	2	38	divisor[7]	quotient[1]	2.479	1.388	1.092	-∞	input port clock			0.000
↳ Path 12	∞	3	2	38	divisor[7]	quotient[5]	2.504	1.372	1.132	-∞	input port clock			0.000
↳ Path 13	∞	3	2	38	divisor[7]	quotient[2]	2.564	1.365	1.200	-∞	input port clock			0.000
↳ Path 14	∞	3	2	38	divisor[7]	quotient[4]	2.570	1.373	1.197	-∞	input port clock			0.000
↳ Path 15	∞	3	2	38	divisor[7]	quotient[6]	2.608	1.383	1.224	-∞	input port clock			0.000
↳ Path 16	∞	4	3	38	divisor[7]	quotient[7]	2.897	1.497	1.400	-∞	input port clock			0.000

4.2.2 Utilization Report

Name	Slice LUTs (134600)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (400)
N divider	290	83	290	27

4.2.3 Power Report

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	13.312 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	49.9°C
Thermal Margin	35.1°C (18.5 W)
Ambient Temperature	25.0 °C
Effective θ_{JA}	1.9°C/W
Power supplied to off-chip devices	0 W
Confidence level	Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power

Dynamic:	13.117 W	(99%)
26% Signals:	3.446 W	(26%)
33% Logic:	4.310 W	(33%)
41% I/O:	5.361 W	(41%)
Device Static:	0.195 W	(1%)

5 Goldschmidt Divider

5.1 Synthesis Reports

5.1.1 Timing Summary

Setup Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 1	∞	126	127	61	clk	quotient[4]	81.272	16.555	64.717	∞	input port clock			0.000
 Path 2	∞	126	127	61	clk	quotient[5]	81.272	16.555	64.717	∞	input port clock			0.000
 Path 3	∞	126	127	61	clk	quotient[0]	81.271	16.535	64.736	∞	input port clock			0.000
 Path 4	∞	126	127	61	clk	quotient[1]	81.271	16.535	64.736	∞	input port clock			0.000
 Path 5	∞	125	126	61	clk	quotient[2]	81.211	16.430	64.781	∞	input port clock			0.000
 Path 6	∞	125	126	61	clk	quotient[6]	81.211	16.430	64.781	∞	input port clock			0.000
 Path 7	∞	125	126	61	clk	quotient[3]	81.074	16.430	64.644	∞	input port clock			0.000
 Path 8	∞	127	128	61	clk	quotient[7]	81.061	16.660	64.401	∞	input port clock			0.000

Hold Time:

Name	Slack 	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
 Path 9	∞	3	4	37	divisor[7]	quotient[4]	1.993	1.353	0.640	-∞	input port clock			0.000
 Path 10	∞	3	4	37	divisor[7]	quotient[5]	1.993	1.353	0.640	-∞	input port clock			0.000
 Path 11	∞	3	4	37	divisor[7]	quotient[0]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 12	∞	3	4	37	divisor[7]	quotient[1]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 13	∞	3	4	37	divisor[7]	quotient[2]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 14	∞	3	4	37	divisor[7]	quotient[3]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 15	∞	3	4	37	divisor[7]	quotient[6]	1.994	1.354	0.640	-∞	input port clock			0.000
 Path 16	∞	3	4	11	divisor[3]	quotient[7]	1.994	1.354	0.640	-∞	input port clock			0.000

5.1.2 Utilization Report

Name 	Slice LUTs (134600)	F7 Muxes (67300)	Bonded IOB (400)
 gdivider	323	1	27

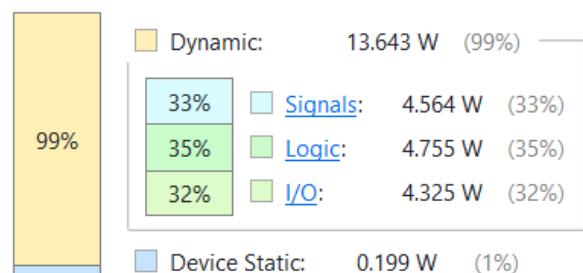
5.1.3 Power Report

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power:	13.842 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	50.9°C
Thermal Margin	34.1°C (18.0 W)
Ambient Temperature	25.0 °C
Effective θ_{JA}	1.9°C/W
Power supplied to off-chip devices	0 W
Confidence level	Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power



5.2 Implementation Reports

5.2.1 Timing Summary

Setup Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 1	∞	128	126	61	clk	quotient[0]	97.286	18.112	79.174	∞	input port clock			0.000
↳ Path 2	∞	128	126	61	clk	quotient[4]	97.116	17.913	79.204	∞	input port clock			0.000
↳ Path 3	∞	128	126	61	clk	quotient[6]	97.066	17.923	79.143	∞	input port clock			0.000
↳ Path 4	∞	128	126	61	clk	quotient[5]	96.963	18.096	78.867	∞	input port clock			0.000
↳ Path 5	∞	128	126	61	clk	quotient[7]	96.741	17.923	78.818	∞	input port clock			0.000
↳ Path 6	∞	128	126	61	clk	quotient[2]	96.664	17.904	78.760	∞	input port clock			0.000
↳ Path 7	∞	128	126	61	clk	quotient[1]	96.653	17.928	78.725	∞	input port clock			0.000
↳ Path 8	∞	128	126	61	clk	quotient[3]	96.648	17.899	78.749	∞	input port clock			0.000

Hold Time:

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 9	∞	3	2	11	divisor[3]	quotient[7]	2.433	1.389	1.044	-∞	input port clock			0.000
↳ Path 10	∞	3	2	37	divisor[7]	quotient[2]	2.513	1.365	1.149	-∞	input port clock			0.000
↳ Path 11	∞	3	2	37	divisor[7]	quotient[1]	2.515	1.388	1.127	-∞	input port clock			0.000
↳ Path 12	∞	3	2	37	divisor[7]	quotient[3]	2.517	1.360	1.157	-∞	input port clock			0.000
↳ Path 13	∞	3	2	37	divisor[7]	quotient[4]	2.597	1.373	1.224	-∞	input port clock			0.000
↳ Path 14	∞	3	2	37	divisor[7]	quotient[5]	2.616	1.437	1.178	-∞	input port clock			0.000
↳ Path 15	∞	3	2	37	divisor[7]	quotient[6]	2.634	1.383	1.250	-∞	input port clock			0.000
↳ Path 16	∞	3	2	37	divisor[7]	quotient[0]	2.670	1.448	1.222	-∞	input port clock			0.000

5.2.2 Utilization Report

Name	Slice LUTs (134600)	F7 Muxes (67300)	Slice (33650)	LUT as Logic (134600)	Bonded IOB (400)
gdivider	321	1	88	321	27

5.2.3 Power Report

Summary

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:	13.263 W
Design Power Budget:	Not Specified
Process:	typical
Power Budget Margin:	N/A
Junction Temperature:	49.8°C
Thermal Margin	35.2°C (18.6 W)
Ambient Temperature	25.0 °C
Effective θ_{JA}	1.9°C/W
Power supplied to off-chip devices	0 W
Confidence level	Low
Launch Power Constraint Advisor to find and fix invalid switching activity	

On-Chip Power

